

RealityCrafter: User-guided Editable 3D Scene Generation from a Single Image in Mixed Reality

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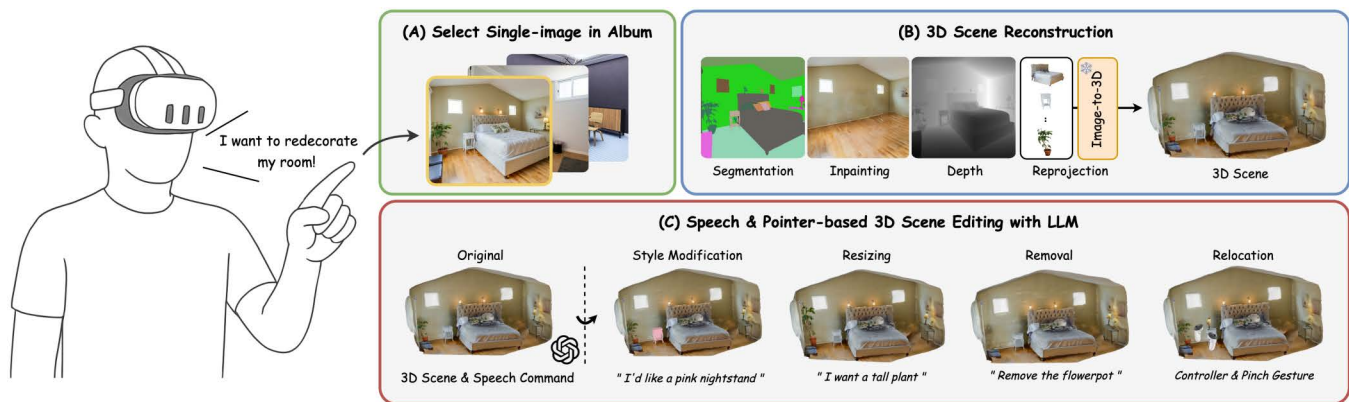


Figure 1: RealityCrafter creates an editable 3D scene driven by user speech from a single real-world image. Our system (a) takes a single image as input; (b) reconstructs it into a 3D scene; and (c) empowers users to author that space via voice commands and pointer interactions.

Abstract

We propose RealityCrafter, a mixed-reality 3D authoring tool that enables users to edit and interact with a reconstructed 3D scene from a single real-world image. Prior research has largely focused on 3D authoring tools for purely virtual spaces, insufficiently incorporating real-world context and thereby hindering user immersion during the creation process. To overcome these limitations, our approach takes a single real-world image as input, generates segmented object-level 3D meshes in a zero-shot manner, and reconstructs a 3D scene where objects can be removed or modified without occlusion through instance mask-based inpainting. We leverage LLMs to interpret user voice commands and update the style, position, scale, and orientation of 3D objects in real time,

providing an interactive 3D authoring interface in mixed-reality environments. By using a single image as a baseline, this approach enables effortless generation of realistic 3D scenes and intuitive editing based on user intent, delivering a novel creative experience that seamlessly blends the real and the virtual objects.

CCS Concepts

• Human-centered computing → Interactive systems and tools; 3D Authoring; • Computing methodologies → Mixed reality; Computer vision.

Keywords

Graphics; Mixed Reality; Generative AI; AI assisted creativity tool

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1 Introduction

With the recent significance of spatial computing technologies to extend real-world spaces into mixed reality (MR), there is a growing need for authoring tools that allow users to create immersive 3D content directly based on their physical environment [2, 8]. Furthermore, as MR systems increasingly incorporate speech, text, and pointers, interfaces for multi-modal intent inference and adaptive 3D manipulation have become essential [5].

While many studies have explored 3D scene authoring in AR/VR, systems enabling real-world grounded editing and seamless virtual blending remain challenging. [14] introduced generative models into the VR authoring pipeline for AI-user collaborative layout creation, but its reliance on synthetic datasets limited its applicability to real settings. Similarly, [11] adopted a radiance field-based approach for photorealistic editing, yet its functionality was constrained to pre-constructed virtual scenes.

In this paper, we present RealityCrafter, a MR authoring system that reconstructs a 3D scene from a single real-world image and enables user-guided editing and customization. We incorporate amodal completion [12] and object mask-based inpainting [3], allowing occluded regions to be plausibly reconstructed and removed objects to blend seamlessly with the background. User speech is parsed via a large language models (LLMs), enabling intuitive control over object style, scale and position. This allows users to generate and customize an editable 3D scene from a single image of a past, current, or imagined space through an interactive interface.

2 System Implementation

RealityCrafter is an interface system designed to generate and edit 3D scenes from a single real-world image. The overall system consists of two main stages: (1) offline 3D scene reconstruction from a single image, and (2) online user-guided 3D object editing and creative authoring.

2.1 3D Reconstruction from a Single Image

Robust Image Understanding in the Wild. The core challenge of reconstructing 3D scenes from a single image lies in capturing rich features despite occluded or non-visible regions. To address this, we estimate depth [9], camera parameters [6], and semantic segmentation [13] from the input image to recover both the global layout and instance-level information of the scene. Inspired by [4], we adopt a pretrained diffusion model [12] to perform amodal completion, filling in the occluded parts of each object instance based on segmented crop image. This pipeline enables robust object-level feature extraction from real-world images captured in the wild, and generalizes to multi-object scenes in a zero-shot manner.

3D Scene and Background Generation. RealityCrafter constructs the 3D scene by handling object instances and background separately. Objects are individually reconstructed using a single image-to-3D module [7] and reprojected in the scene according to the predicted depth map. To generate the background, we apply image inpainting [3] guided by segmented object masks, ensuring visual consistency even after object removal. The inpainted background is rendered as a continuous surface using an SDF representation

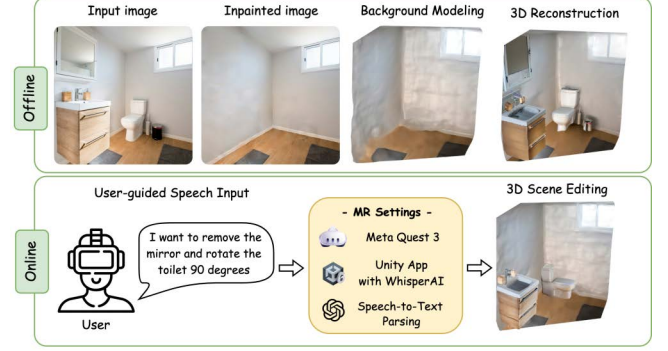


Figure 2: Our system reconstructs a 3D scene from a single image in an offline stage, then moves into an online phase where users direct object edits and craft 3D content.

derived from a depth-based point cloud. Finally, object instances and background are merged as meshes into a unified 3D scene that preserves the original camera viewpoint.

2.2 Interactive Creation of Editable 3D Scenes

LLM-driven Speech Command Parsing. The user’s voice commands are transcribed into text in real time using Whisper AI [1] and are treated as interaction inputs for the reconstructed 3D scene. The transcribed text is passed to the GPT API along with a pre-defined prompt template to extract the command intent and key parameters (e.g., object name, manipulation type and scale). The resulting structured command is then mapped to an internal interpreter, which translates it into concrete actions such as style modification, movement, scaling, or deletion of 3D objects. To ensure reliability, we compute a confidence score by averaging the softmax probabilities of relevant GPT tokens, and request user confirmation when the score falls below a 0.8 threshold to prevent execution errors.

Key Interactions. RealityCrafter seamlessly fuses reconstructed real-world spaces with virtual objects in MR environments and enables intuitive 3D scene authoring through four key interactions:

- **Style modification:** When a speech command like “Make this chair pink” is received, the LLM extracts the style attribute and communicates it to the Python server, which updates the corresponding texture using the InTex [10].
- **Resize & Rotate:** Users can adjust the scale and rotation of real-world-based virtual objects using speech commands.
- **Object removal:** Upon receiving a command like “Delete the table”, the system removes the corresponding virtual object and blends the background naturally with the surrounding.
- **Object relocation:** By dragging and dropping via a VR controller, users can move virtual objects to desired positions.

By combining voice-based control with pointer input, this multi-modal interaction effectively integrates real and virtual content in MR environments and enables users to carry out complex 3D editing tasks intuitively.

3 Potential Applications

RealityCrafter enables users to revive spaces connected to past memories and create interactive 3D content using voice and pointer-based editing in MR environments. In VR interior design, users can reconstruct their living spaces, simulate furniture placement in real time, and explore personalized layouts. It can also be extended into a remote collaboration platform, supporting real-time feedback and co-editing in a shared MR space.

4 Conclusion and Future Work

This paper describes RealityCrafter, a 3D authoring system that combines zero-shot 3D scene reconstruction from a single image with user-guided voice command parsing. The proposed approach enables intuitive and immersive 3D editing in mixed reality by immediately reflecting user intent through voice and pointer-based interactions. In future work, we plan to evaluate the system through user studies and deploy it across real-world application scenarios, with the goal of extending it into a collaborative authoring tool for multi-user MR environments.

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